BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 226 · Tree traversal: post-order and breadth-first

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a constructor do?

Q2. What is in-order traversal order?

Q3. What does FIFO stand for?

Q4. A root R has only left child L. What is post-order?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

post-order — Traverse left, right, then root.

breadth-first — Visit vertices level by level.

FIFO queue — First-in-first-out structure used by BFS.

leaf — Node without children.

traversal — Systematically visit each node in a structure.

New learning

Depth-first post-order — Visit an entire left subtree, then right subtree, and finally the current node.

Breadth-first level order — Start with the root in a queue. Dequeue one node, visit it and enqueue each of its children.

Compare output — Same structure yields different orders: 2,6,4,12,8 versus 8,4,12,2,6.

Worked example

Find post-order traversal of root 10, children 5 and 15, with 5 having left child 2.

1. Predict first — Visit 5's left child 2 first, then node 5.

2. Model the reasoning — Visit right child 15, then root 10. Output 2,5,15,10.

We Do · Practise with guidance

Find breadth-first traversal of root 20, children 10 and 30, with 10 having right child 15.

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Use a FIFO queue: dequeue, visit, enqueue children.
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Predict first

Visit root 20, enqueue 10 and 30.

Model the reasoning

Visit 10 and enqueue child 15, then visit 30; final 15 gives 20,10,30,15.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Root 6 has left 3 and right 9. Give post-order.

Question 2

Tree root 12, left 5 with children 2 and 8, right 17. Give breadth-first and post-order traversals.

Question 3

Write pseudocode to traverse a binary tree breadth-first. Explain why a queue rather than a stack is used.

Exit Ticket · Answer before checking

Breadth-first traversal normally uses a stack to ensure earliest-discovered nodes are visited first.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Initialises an object when it is created.

Q2. Left, root, right.

Q3. First In, First Out.

Q4. L then R.

You Do pages · Independent questions

Q1. [3] Visit left 3 [1], right 9 [1], root 6 [1]; 3,9,6.

Q2. [5] Breadth-first 12,5,17,2,8 [2]; post-order 2,8,5,17,12 [3].

Q3. [6] If root is not null, enqueue(root) [1]; while queue nonempty [1], dequeue and visit [1], enqueue non-null left then right children [2]; queue maintains FIFO order across levels [1].

Exit Ticket · Model answer

False. It uses a FIFO queue.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: